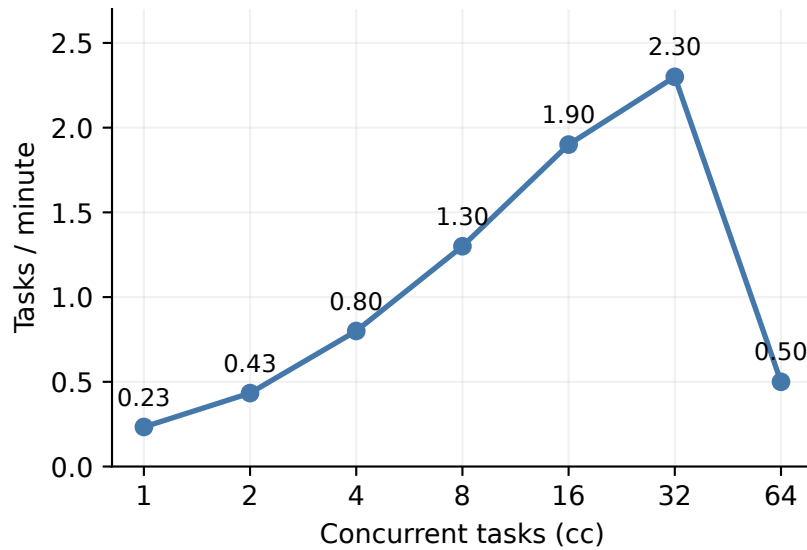
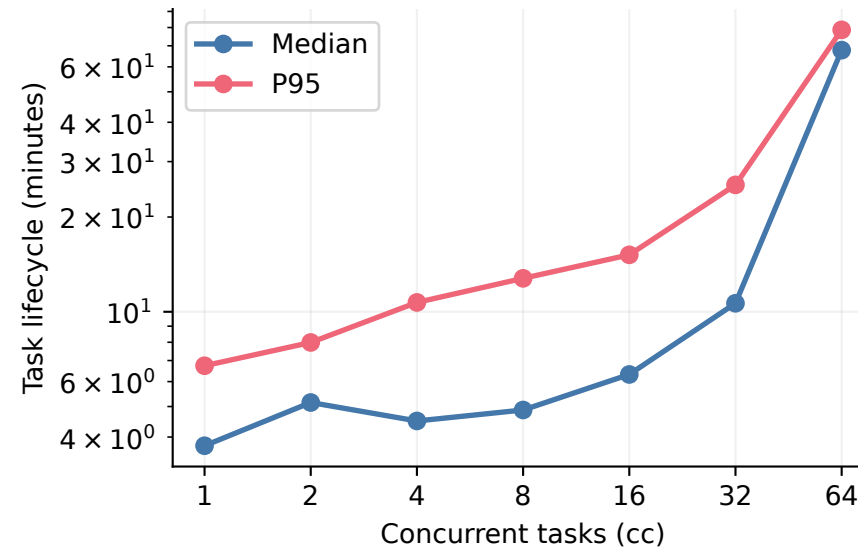


mini-SWE scaling | Qwen3.6-35B-A3B, TP4, 61-trace pool

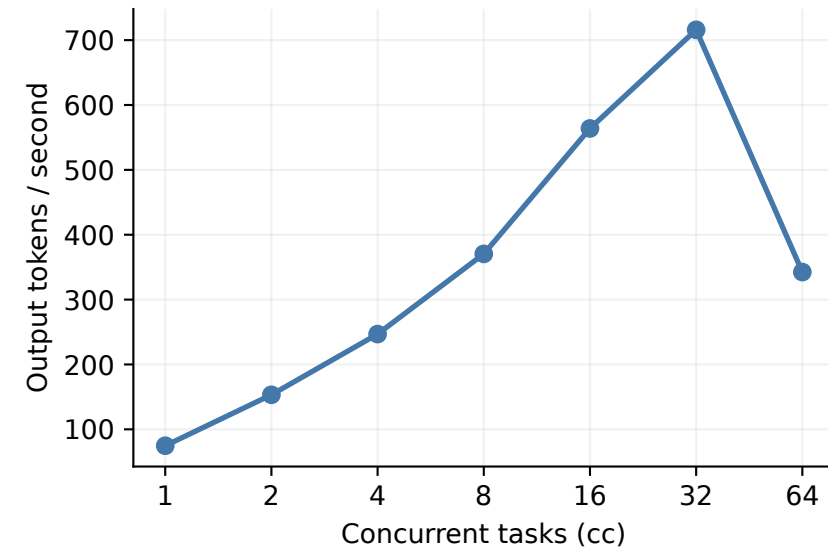
(a) Task throughput



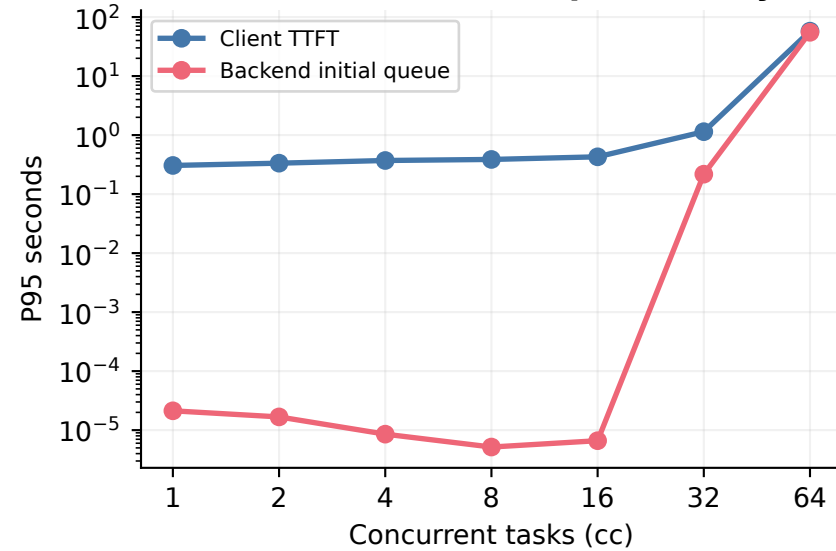
(b) Task latency



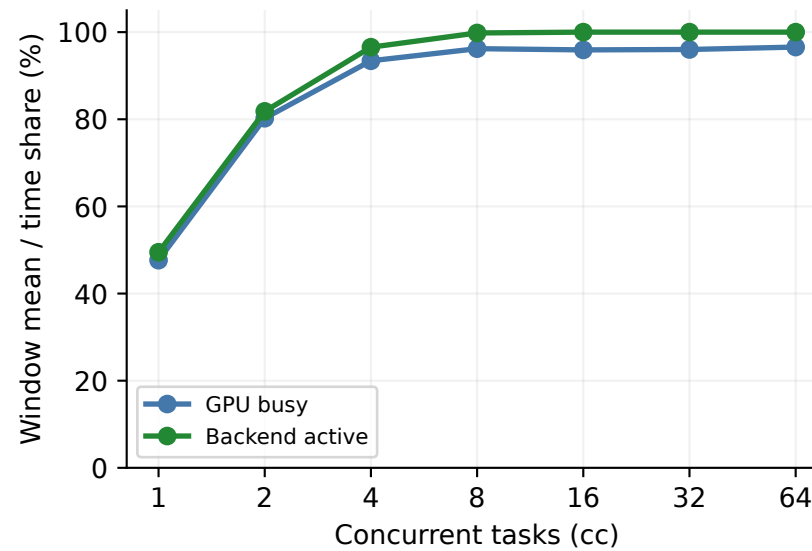
(c) Inference output throughput



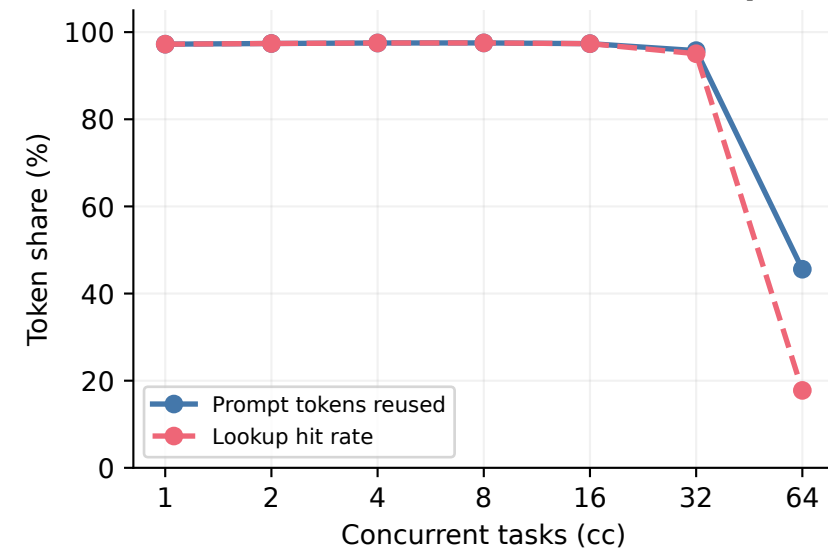
(d) First-token and queue delay



(e) Inference utilization



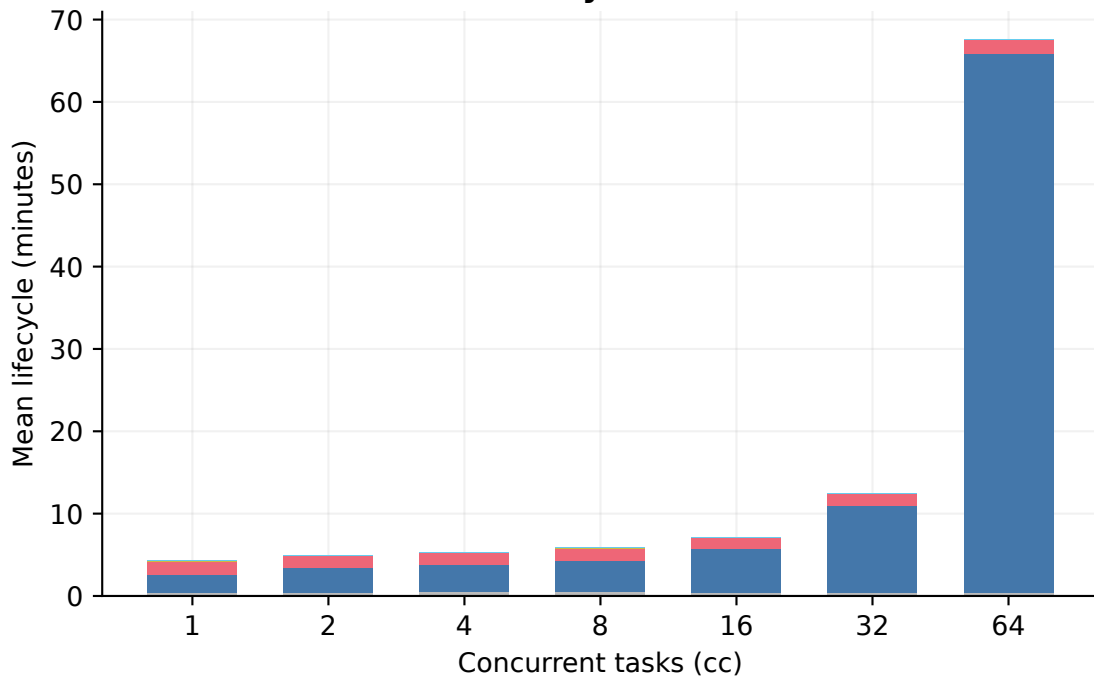
(f) Cache reuse vs. cache lookups



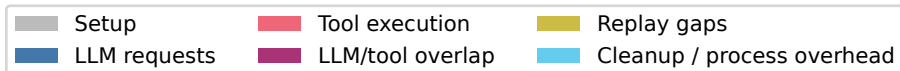
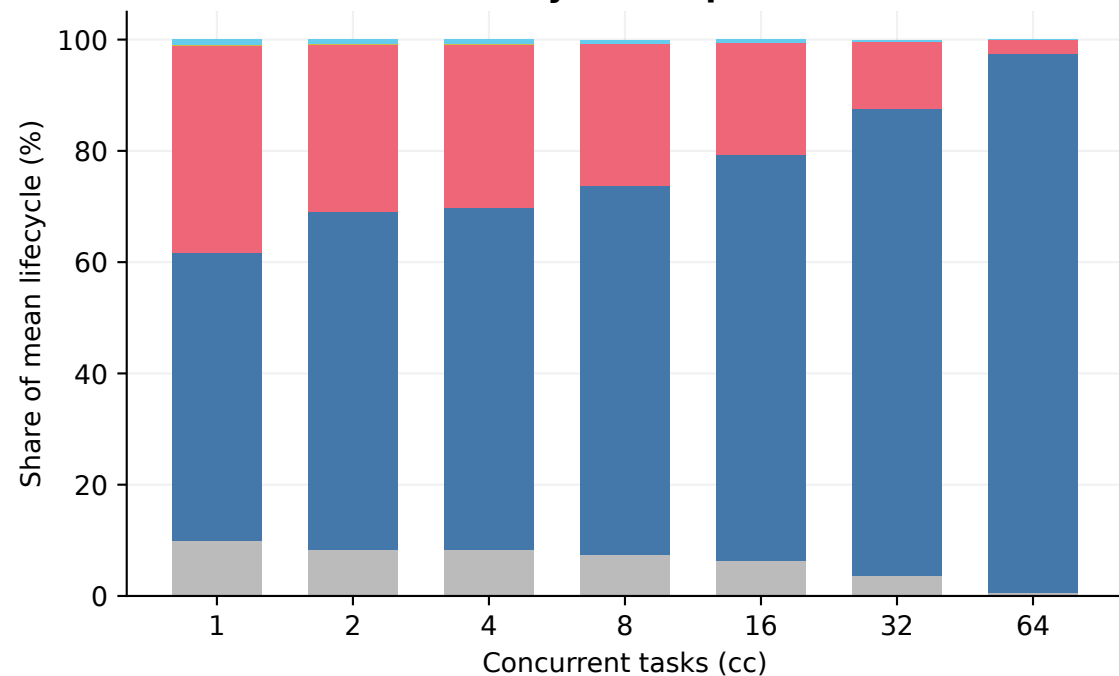
120 s warmup + 1800 s measurement; one repetition per cc. Latencies follow window admissions through natural drain.
Native bash tools; cc32 has the highest observed throughput. cc64 completions all started in warmup.

Where task wall time is spent | disjoint client intervals

(a) Mean lifecycle breakdown



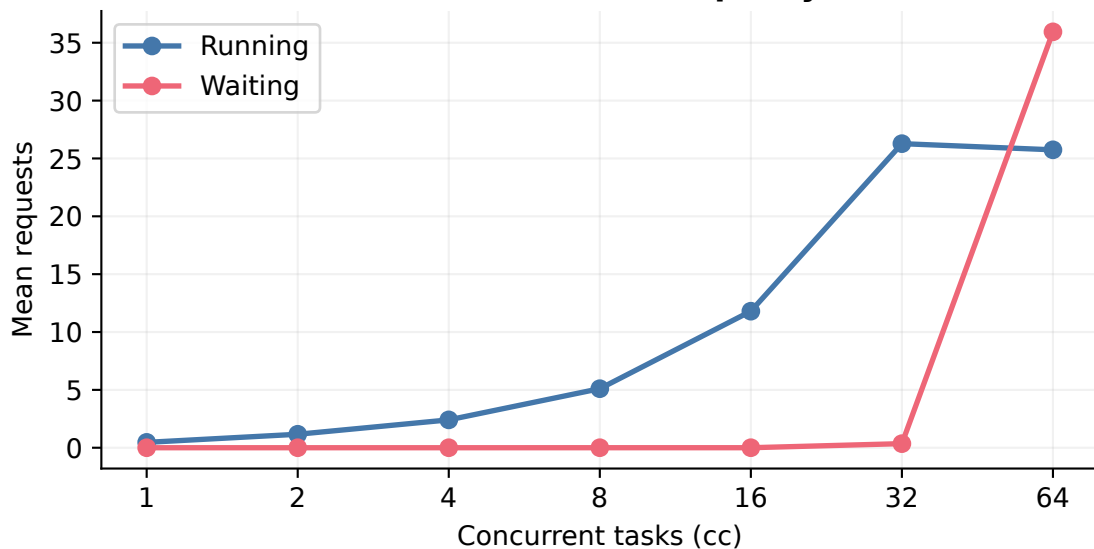
(b) Lifecycle composition



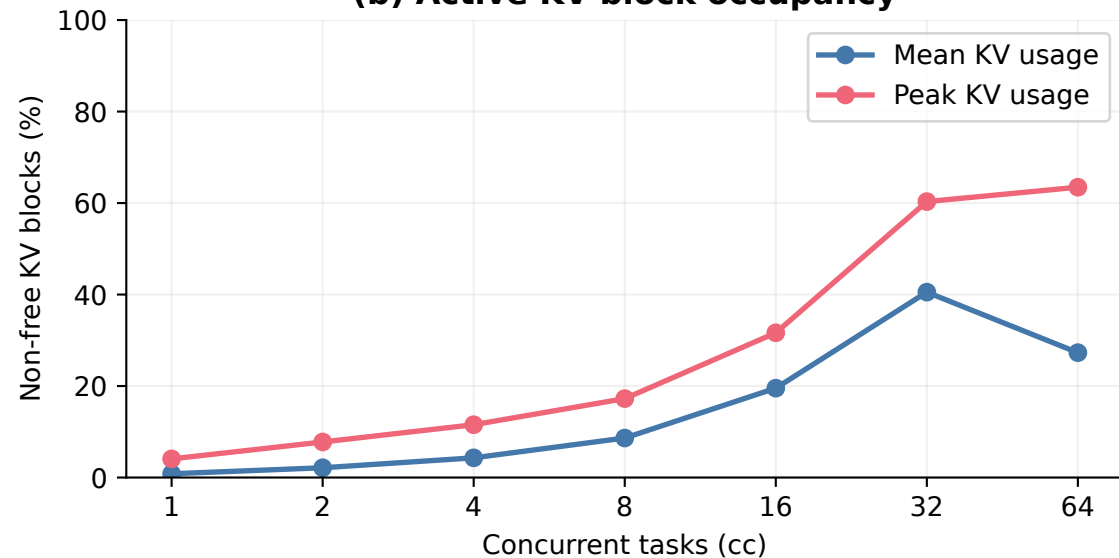
LLM intervals include server waiting and network time. Parallel calls are counted by interval union; overlap is separate.

Backend scheduling, cache and output streaming

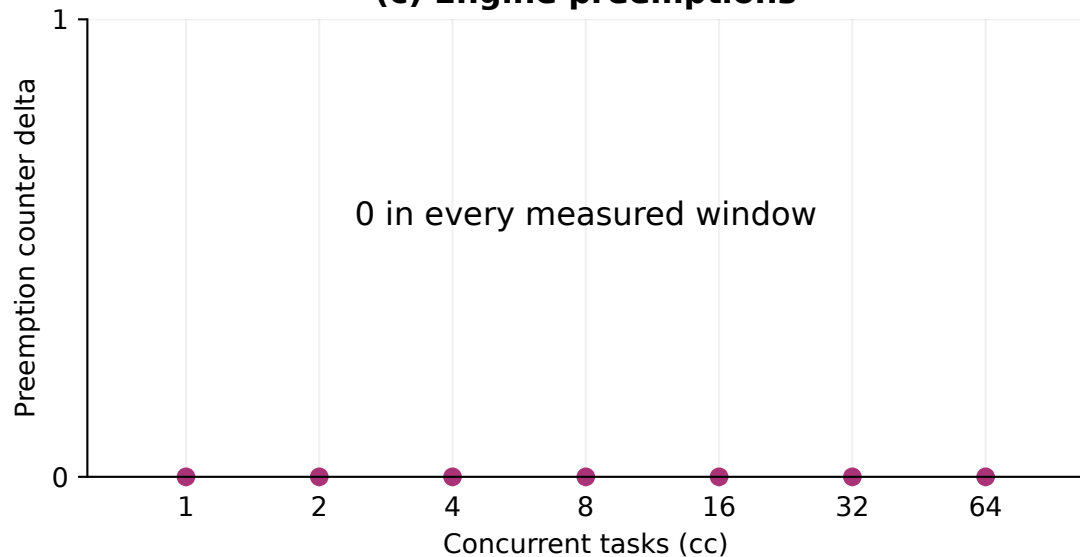
(a) Scheduler occupancy



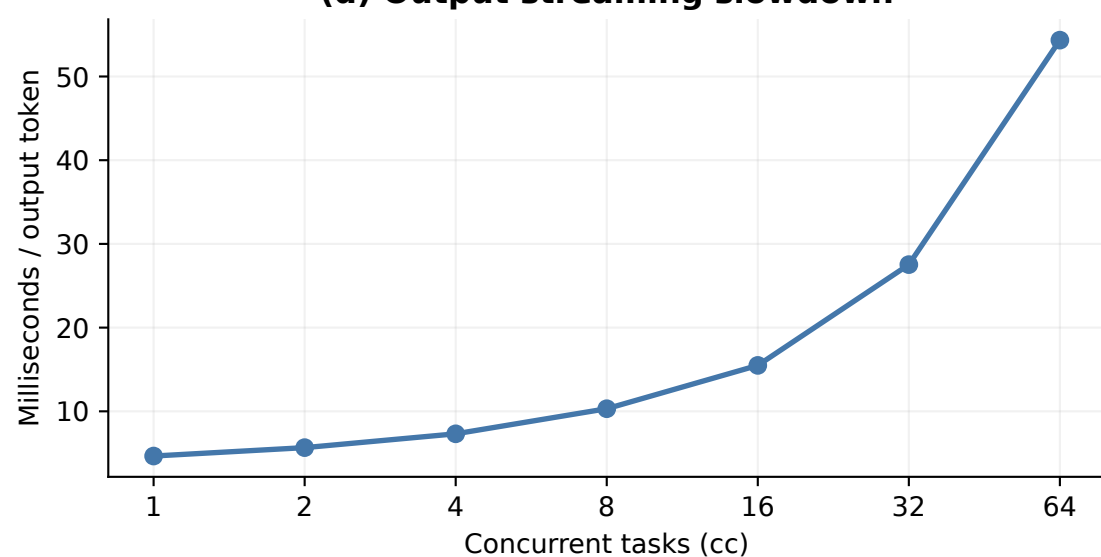
(b) Active KV-block occupancy



(c) Engine preemptions



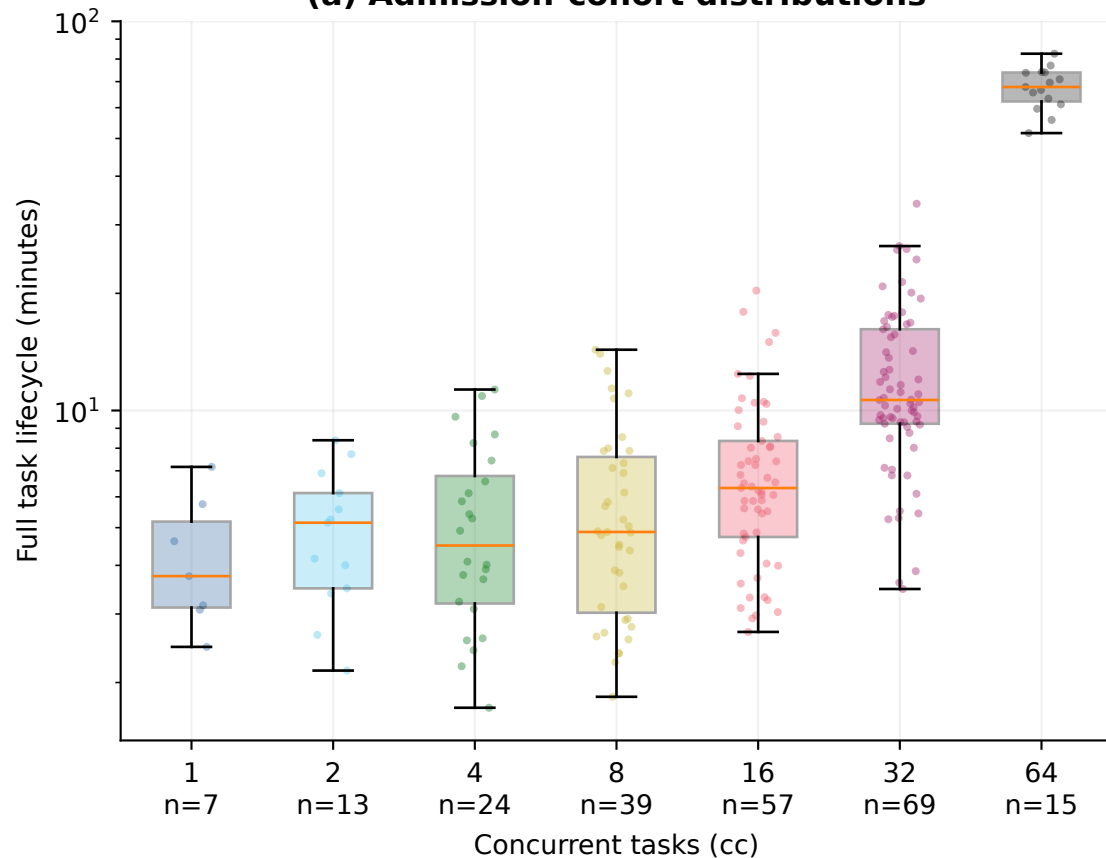
(d) Output streaming slowdown



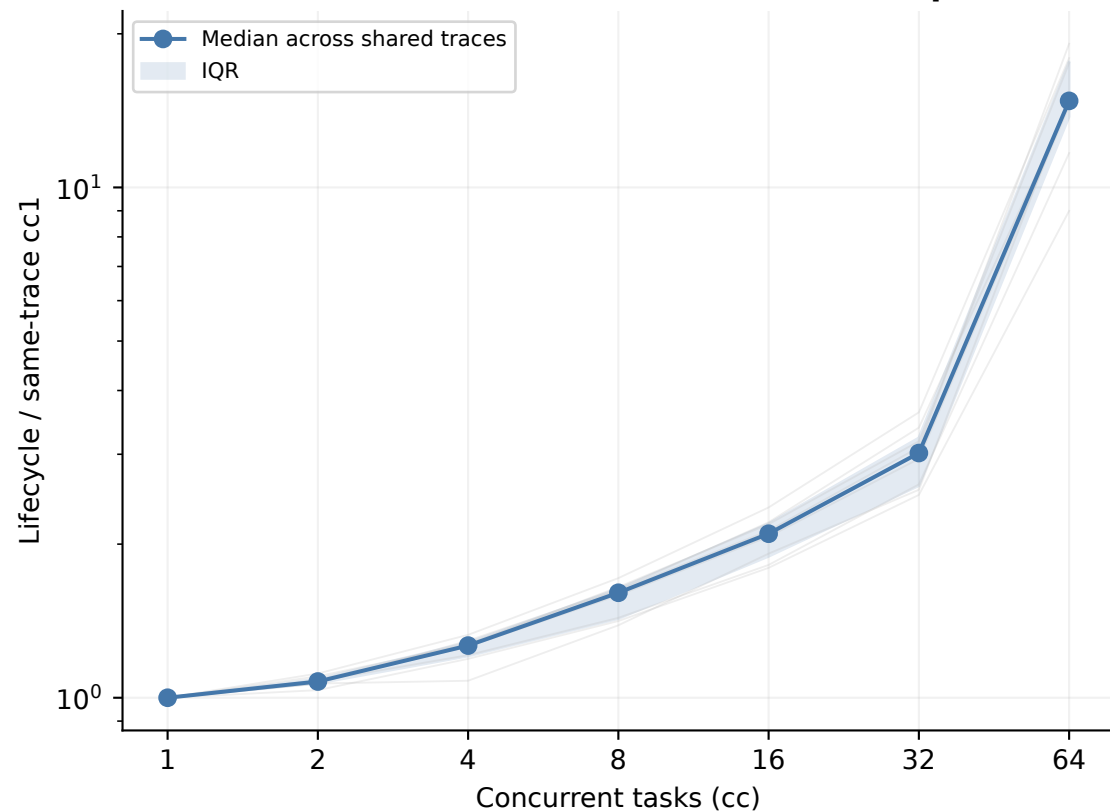
Free KV blocks can still hold evictable cached prefixes; this gauge does not measure all resident prefix data.
Counters span ~1799 s. TPOT is estimated from client streaming chunks, not exact engine per-token timestamps.

Latency distributions and workload-mix sensitivity

(a) Admission-cohort distributions



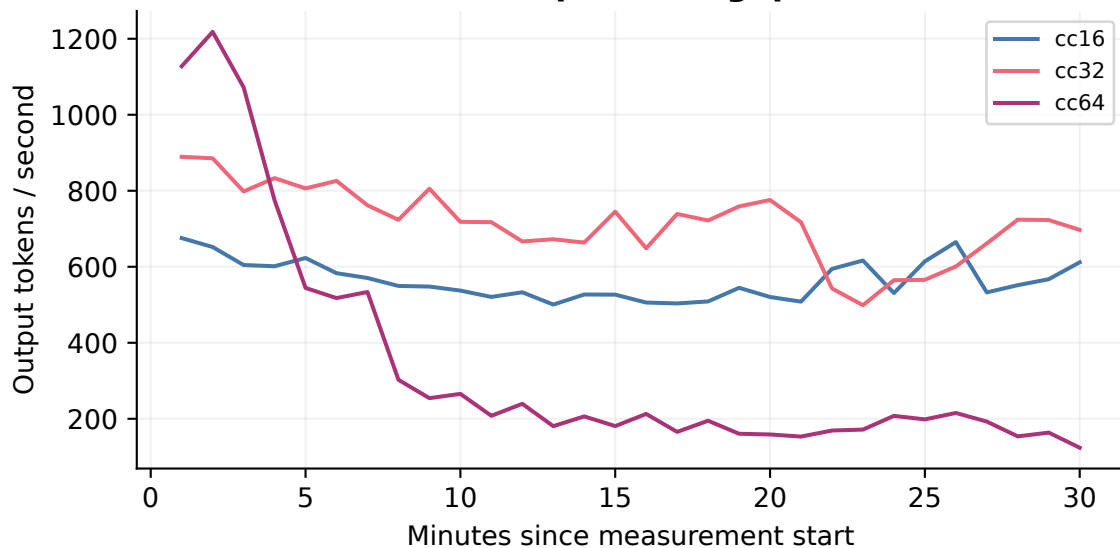
(b) First 8 launches (includes warmup)



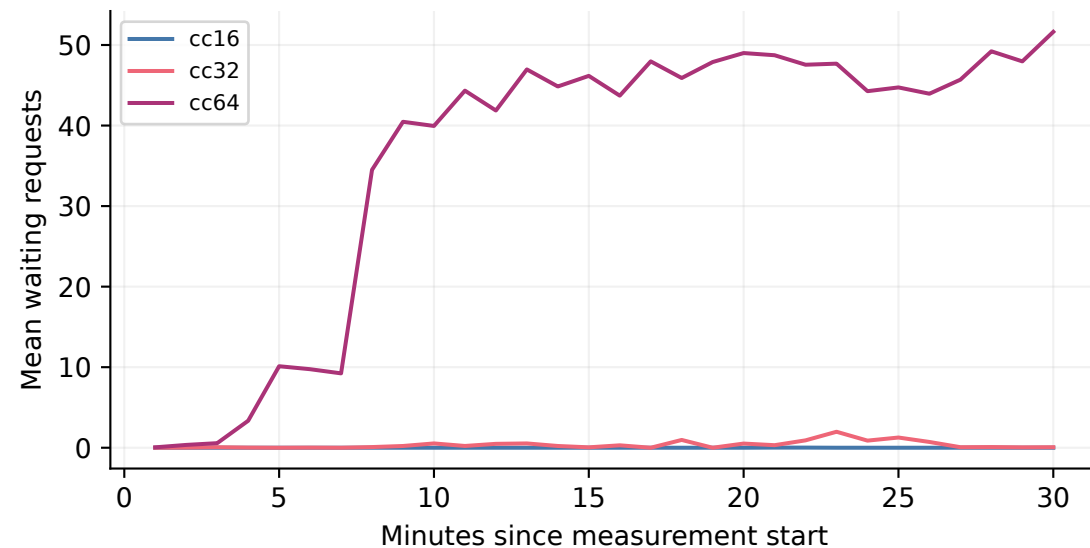
Boxes: median / IQR, whiskers 1.5 IQR; every observed task is shown. No censored task latency is dropped.
Right: same first 8 tasks, one launch each; mixed warmup/measurement phases. Sensitivity check, not steady-state latency.

Within-window trends |cc16, cc32 and cc64

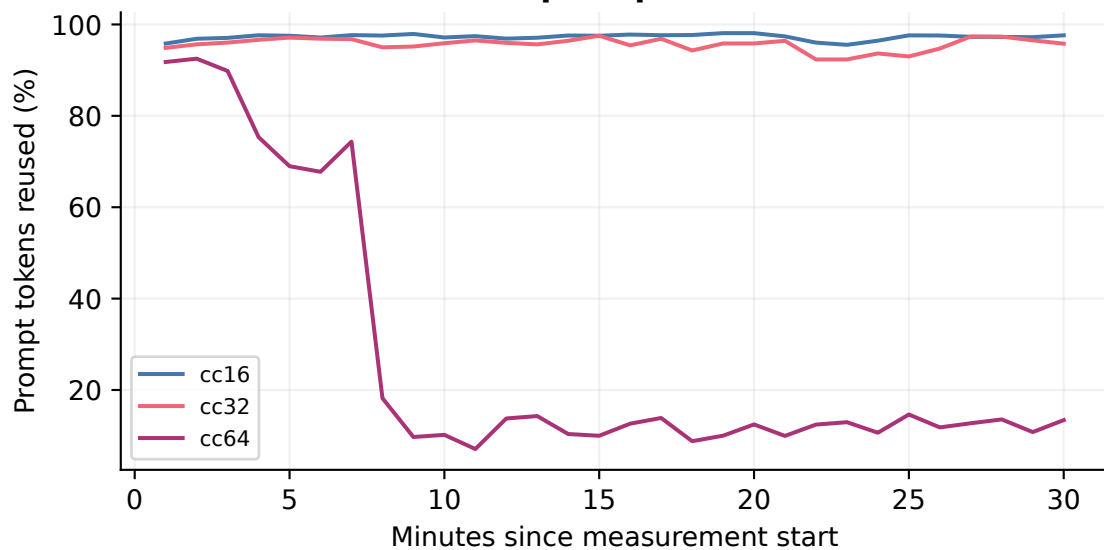
(a) Output throughput



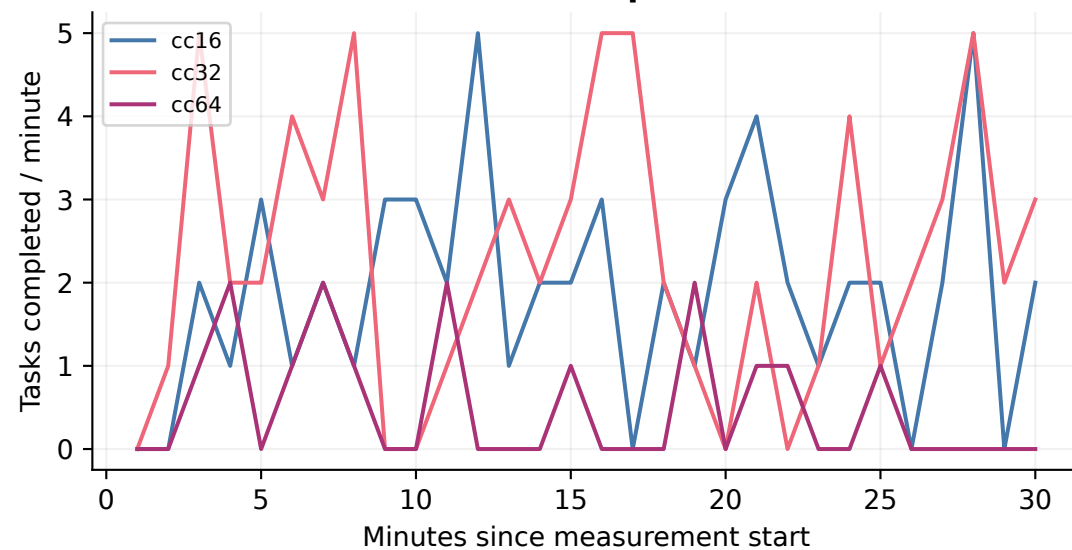
(b) Queue accumulation



(c) Interval prompt-token reuse



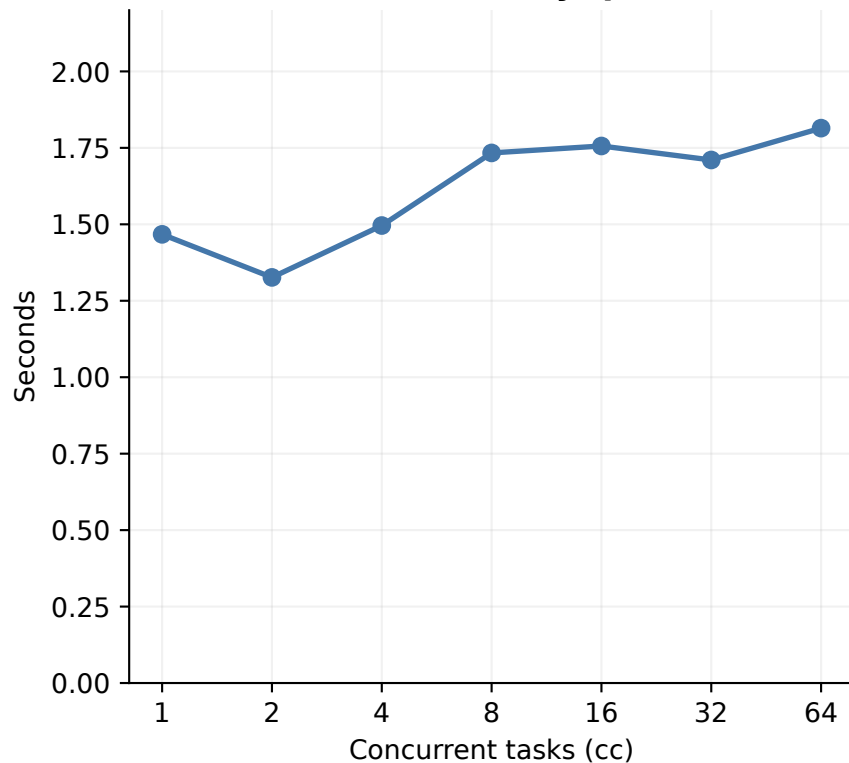
(d) Task completions



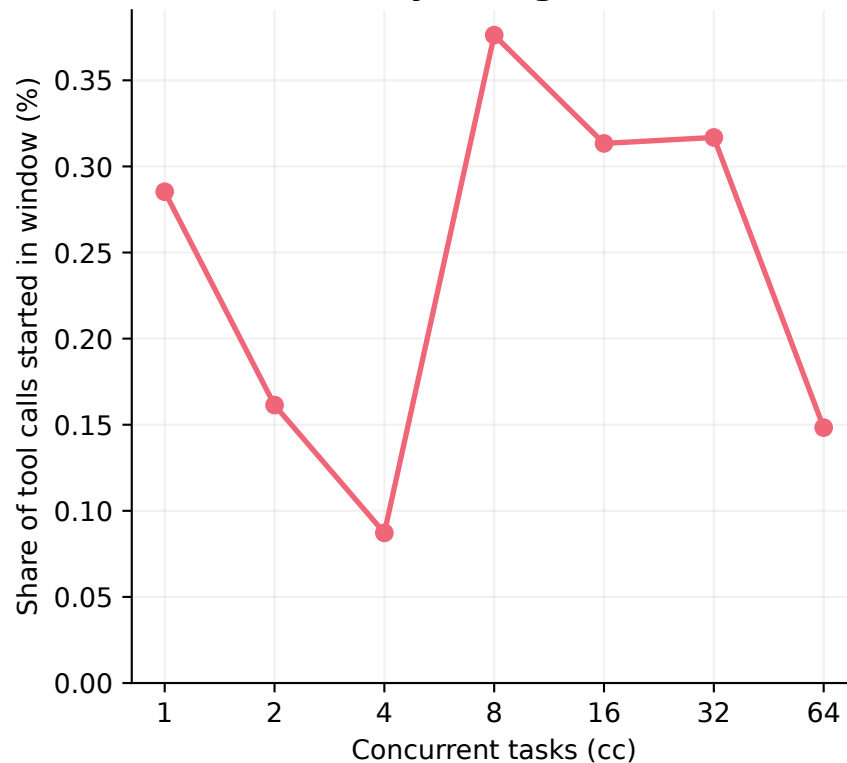
60-second bins; token/cache rates use sampled counter deltas within each bin (~59 s). Task counts use full 60 s bins.
These curves assess transient behavior; a 30-minute measurement does not by itself establish steady state.

Tool behavior and sample coverage

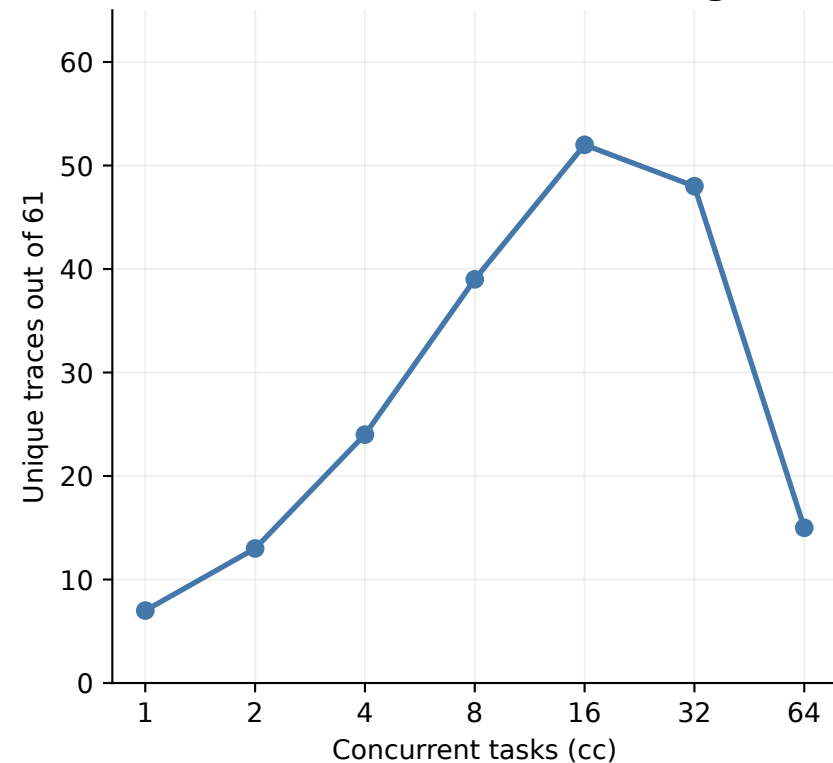
(a) Tool latency (p95)



(b) Newly failing tool calls

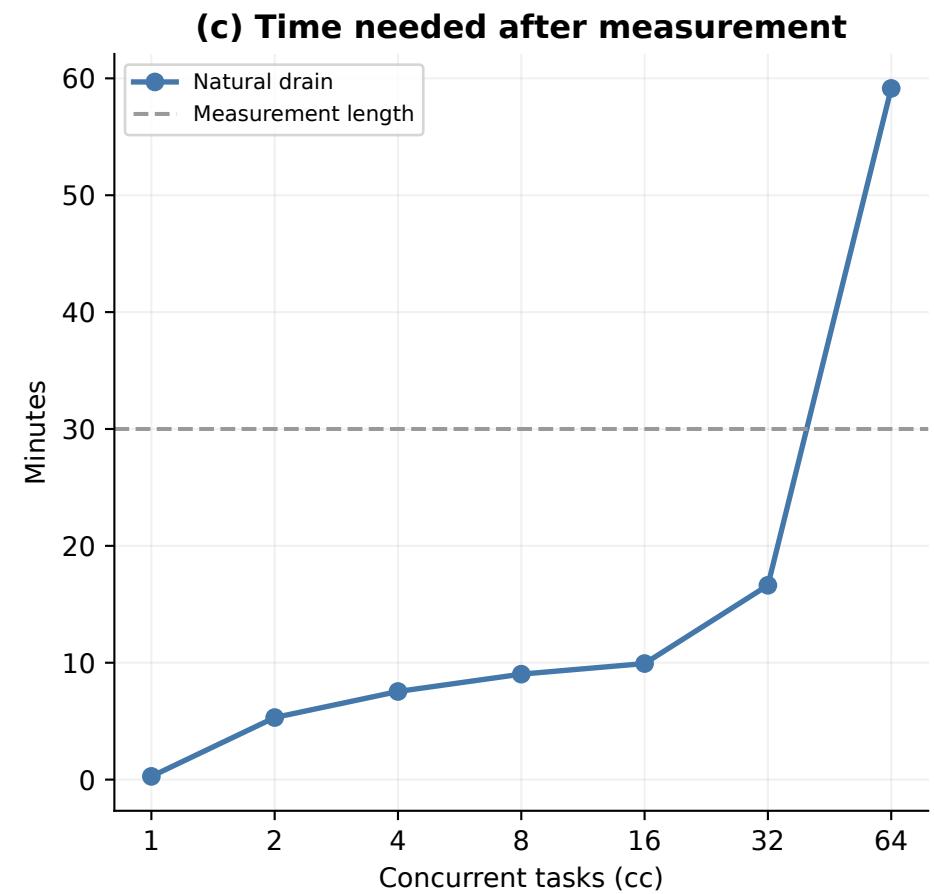
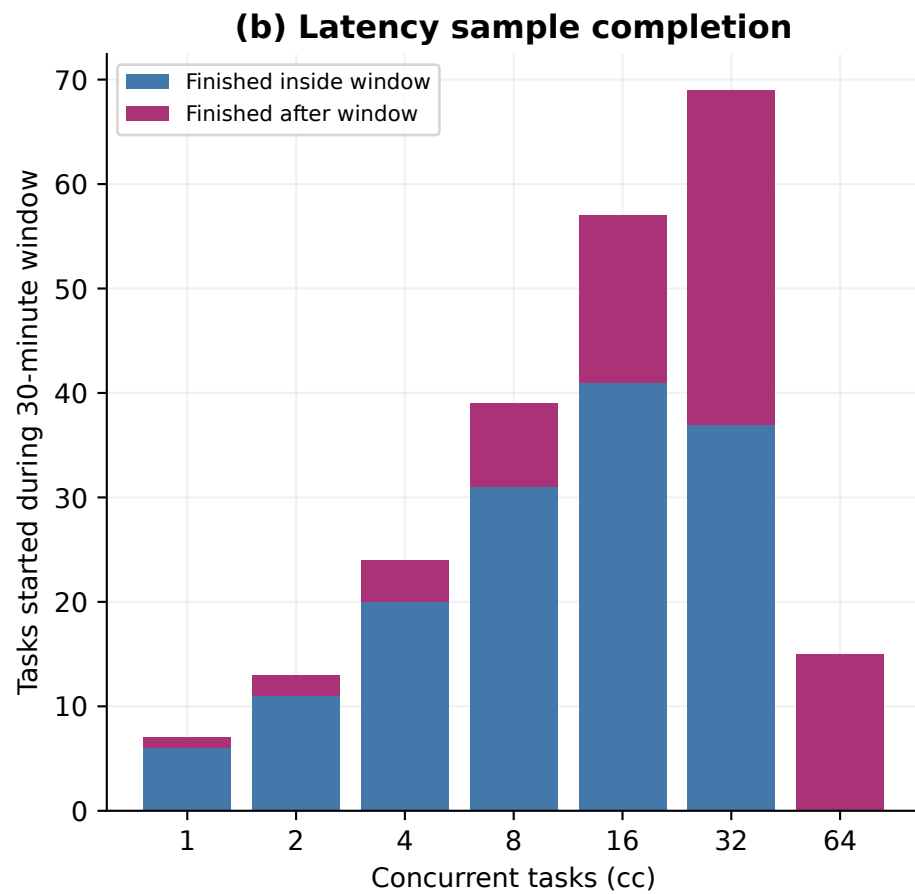
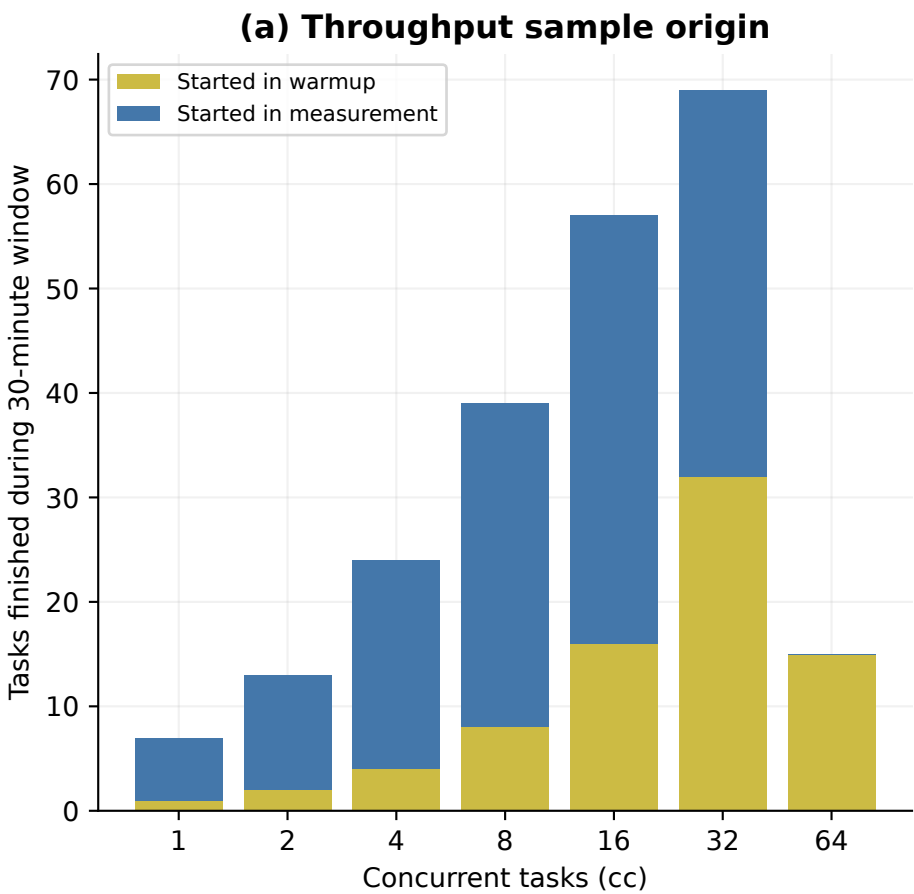


(c) Admission-cohort coverage



Native tool failures are retained; engine validation does not certify identical tool outcomes.
All points use the same trace pool and seed, but complete different task cohorts within a fixed window.

Finite-window effects | keep completion and admission cohorts distinct



cc64: all 15 window completions started in warmup; all 15 window admissions finished later.
All admission-cohort latencies are observed through natural drain; no unfinished task is silently discarded.